A tree condition forces the Shore-blocking functional to be partial

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Editorial note. The counted unit is the article's own Lemma lem:wkl-wf-forcing-wf at source lines 683-685 together with the proof environment that begins two lines after the statement ends (687-709); the pairing is the ordinary adjacent one.

Editorial note. The retained context at source lines 411-413 and 426-431 is the statement of the two earlier lemmas that the counted proof invokes through the author Cref calls, namely the transfer of well-foundedness to the natural product and the Shore-blocking functional; both are retained as prerequisites, their proofs are not retained. The retained context at source line 663 is the author explanation of the forcing relation for a decided initial segment of the jump.

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, his natural-operation macros at source lines 104-107 and his quotation macro at source line 116, and it retains mathabx from source line 103 and the cancel package from source line 9, the latter because the retained proof prints a negated satisfaction sign. The LS2 font-encoding block at source lines 93-99 is not retained because the host TeX Live has no stix directory and the author never uses that symbol. The inputenc load with the utf8x option at source line 8 is not retained because it fights the forced xelatex engine, and cleveref is loaded by the retained preamble itself because the assembler only adds that package when it generates navigation stubs.

Editorial note. No retained line contains a citation, so the material carries no bibliography; the cross-references of the retained proof are all resolved inside the retained context.

Editorial note. The article's conservation theorems, its COH lemmas and its SEM section lemmas are not counted here and are not used as premises.

Allowed standard prerequisites (editorial).

- Tree conditions of the forcing used for WKL. In this section a condition is a pair (sigma, T) of a finite binary string and an M-infinite tree, and

an extension (tau, T_2) of (sigma, T_1) requires tau to extend sigma and T_2 to be contained in T_1, so that the conditions form a forcing of tree type; the requirements of this forcing are listed by the author at source lines 640-643.. Fixes the objects of the counted lemma, namely the condition (sigma, T_1), the extension (sigma, T_2) it produces and the generics G and Y.

- The Shore-blocking functional Gamma_k and well-foundedness of the natural product. Gamma_k is the Turing functional constructed by the author in the Shore-blocking lemma, which turns a descending sequence below an ordinal into one below the natural product with k, and the article proves separately that well-foundedness of an ordinal transfers to its natural product with any coefficient; both statements are retained with this card at source lines 426-431 and 411-413.. The counted lemma is stated for exactly this functional and its proof derives a contradiction from the hypothesis that Gamma_k is total, using the transfer of well-foundedness to the natural product.

- Definability classes and M-finiteness in the model. Sets defined by primitive recursive predicates in a real Y are available in the model, and the article uses throughout that M-infinite trees can be cut down by such predicates while staying M-infinite when the corresponding set is M-infinite.. In the first case the proof cuts the tree T_1 down by the primitive Y-recursive predicate that the functional diverges at a fixed n.

Lemma 0.1. *Let $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{WF}(\alpha)$ for α an ordinal less than ϵ_0. Then $\mathcal{M} \models \mathsf{WF}(\alpha \dot{\times} k)$ for every $k \in M$.*

Lemma 0.2. *Consider a model $\mathcal{M} = (M, S)$ and $\alpha \in M$ an ordinal less than ϵ_0. For every $k \in M$, there is a Turing functional Γ_k such that, letting $\beta \in \alpha$ be the largest ordinal less than α of index at most k, for every $X \in 2^M$ such that $\mathcal{M}[X] \models \mathsf{RCA}_0$, if there is some $e < k$ such that Φ_e^X is an M-infinite decreasing sequence of elements smaller than β, then Γ_k^X is an M-infinite decreasing sequence of elements smaller than $\beta \dot{\times} k$.*

Moreover, an index of Γ_k can be found computably in k.

In what follows, $(\tau, U) \Vdash (G \oplus Y)' \upharpoonright_k = \sigma'$ means that for every $e < |\sigma'|$, if $\sigma'(e) = 1$ then $(\tau, U) \Vdash \Phi_e^{G \oplus Y}(e) \downarrow$ and if $\sigma'(e) = 0$, then $(\tau, U) \Vdash \Phi_e^{G \oplus Y}(e) \uparrow$. Thus, this relation is $\Pi^0_1(Y)$.

Lemma 0.3. *Let (σ, T_1) be a condition. For every $k \in M$, letting Γ_k be the functional of Lemma 0.2, there is an extension $(\sigma, T_2) \leqslant (\sigma, T_1)$ forcing $\Gamma_k^{G \oplus Y}$ to be partial.*

Proof. By twisting Γ_k, we can assume that for every $e, n \in M$, if $\Gamma_k^{\sigma \oplus Y}(n) \downarrow$, then

- $\Gamma_k^{\sigma \oplus Y}(m) \downarrow$ for every $m < n$;

- $\Gamma_k^{\sigma\oplus Y}(0), \Gamma_k^{\sigma\oplus Y}(1), \ldots, \Gamma_k^{\sigma\oplus Y}(n)$ is a strictly $<_{\epsilon_0}$-decreasing sequence of ordinals smaller than $\beta \dot{\times} k$.

We have two cases.

Case 1: there exists some $n \in M$ such that for every $\ell \in M$, there is some $\tau \in T_1$ of length ℓ such that $\Gamma_k^{\tau\oplus Y}(n) \uparrow$. Let $T_2 = \{\tau \in T_1 : \Gamma_k^{\tau\oplus Y}(n) \uparrow\}$. Note that the set T_2 is primitive Y-recursive, as the set T_1 and the predicate $\Gamma_k^{\sigma\oplus Y}(n) \uparrow$ are primitive Y-recursive. The set T_2 is closed-downward, and by assumption, is M-infinite. Then $(\sigma, T_2) \leqslant (\sigma, T_1)$. Moreover, by construction, $(\sigma, T_2) \Vdash \Gamma_k^{G\oplus Y}(n) \uparrow$.

Case 2: for every $n \in M$, there is some $\ell_n \in M$ such that for every $\tau \in T$ of length ℓ_n, $\Gamma_k^{\tau\oplus Y}(n) \downarrow$.

Then, for every n, let $\alpha_n = \max\{\Gamma_k^{\tau\oplus Y}(n) : \tau \in T_1 \wedge |\tau| = \ell_n\}$. We claim that for every $n \in M$, $\alpha_{n+1} <_{\epsilon_0} \alpha_n$. Indeed, for every $\tau \in T_1$ such that $|\tau| = \ell_{n+1}$, $\Gamma_k^{\tau\oplus Y}(n+1) <_{\epsilon_0} \Gamma_k^{(\tau\restriction\ell_n)\oplus Y}$, so

$$\max\{\Gamma_k^{\tau\oplus Y}(n+1) : \tau \in T_1 \wedge |\tau| = \ell_{n+1}\} <_{\epsilon_0} \max\{\Gamma_k^{\tau\oplus Y}(n) : \tau \in T_1 \wedge |\tau| = \ell_n\}$$

So $\mathcal{M} \not\models \mathsf{WF}(\alpha \dot{\times} k)$. However, by Lemma 0.1, since $\mathcal{M} \models \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$, $\mathcal{M} \models \mathsf{WF}(\alpha \dot{\times} k)$. Contradiction. □